Solutions and R Programming Examples: Hypothesis Testing_2

1.Weighting change in Anorexic girls

```
aft<-
c(85.2,88.6,86.4,81.9,76.4,93.6,98.4,89.4,71.4,82.1,100.5,95.3,87.6,72.5,78.1,71.3,89.4,92.1,83.9,82.7,81.6,75.7,82.6,
95.4,85.2,83.6,88.6,86.2,86.7)

bef<-
c(80.5,84.9,81.5,82.6,79.9,88.7,94.9,76.3,81.0,80.5,85,89.2,81.3,76.5,70,80.6,83.3,87.7,84.2,86.4,83,76.5,80.2,87.8,8
3.3,79.7,84.5,80.8,87.4)

diff<-aft-bef

boxplot(diff)

shapiro.test(diff)
```

....

```
        Shapiro-Wilk normality test

data:  diff

W = 0.96301, p-value = 0.3889
```

....

t.test(diff, alternative = "greater",conf.level=0.95)

....

One Sample t-test

data: diff

t = 2.4686, df = 28, p-value = 0.009966

alternative hypothesis: true mean is greater than 0

95 percent confidence interval:

 0.7954868   Inf

sample estimates:

mean of x

 2.558621

....

t.test(aft, bef, paired = TRUE, alternative = "greater",con.level=0.95)

....

2. Gum flavor longevity

f<- c(15,21,29,22,19,25,35,23)

m< -c(22,24,23,30,12,17,28)

shapiro.test(f)

....

Shapiro-Wilk normality test

data: f

W = 0.96541, p-value = 0.8597

....

```
shapiro.test(m)
```

....

```
	Shapiro-Wilk normality test

data:  m
W = 0.95878, p-value = 0.8082
```

....

```
var.test (f, m, alternative = "two.sided")
```

....

```
	F test to compare two variances

data:  f and m
F = 0.99331, num df = 7, denom df = 6, p-value = 0.9771
alternative hypothesis: true ratio of variances is not equal to 1
95 percent confidence interval:
 0.1744029    5.0843345
sample estimates:
ratio of variances
         0.9933064
```

....

```
qf(0.975,7,6)
[1] 5.69547
qf(0.025,7,6)
[1] 0.195366
```

....

```
time <- c(15,21,29,22,19,25,35,23,22,24,23,30,12,17,28)

sex <- c(rep("Female",8), rep("Males",7))

m.data<- data.frame(time,sex)

library(car)

leveneTest(time ~ sex, m.data, center=mean)
```

....

```
Levene's Test for Homogeneity of Variance (center = mean)
      Df F value Pr(>F)
group  1       0 0.9997
      13
```

....

```
t.test (f,m,var.equal=TRUE)
```

....

```
        Two Sample t-test

data:  f and m

t = 0.41924, df = 13, p-value = 0.6819

alternative hypothesis: true difference in means is not equal to 0

95 percent confidence interval:

 -5.562215  8.240786

sample estimates:

mean of x          mean of y

 23.62500           22.28571
```

....

3. Advertising claim

L<-c(16.8,11.7,15.6,16.7,17.5,18.1,14.1,21.8,13.9,20.8)

C<-c(22,15.2,18.7,15.6,20.8,19.5,17.0,19.5,16.5,24)

shapiro.test (L)

....

Shapiro-Wilk normality test

data: L

W = 0.97274, p-value = 0.9151

....

shapiro.test(C)

....

Shapiro-Wilk normality test

data: C

W = 0.95772, p-value = 0.7595

....

var.test(L, C, alternative = "two.sided")

....

F test to compare two variances

data: L and C

F = 1.1664, num df = 9, denom df = 9, p-value = 0.8224

alternative hypothesis: true ratio of variances is not equal to 1

95 percent confidence interval:

0.2897208 4.6959767

sample estimates:

ratio of variances

1.166414

....

```
time<-c(16.8,11.7,15.6,16.7,17.5,18.1,14.1,21.8,13.9,20.8, 22,15.2,18.7,15.6,20.8,19.5,17.0,19.5,16.5,24)

location <- c(rep("Local",10), rep("Chain",10))

m.dataframe <- data.frame(time,location)

leveneTest(time ~ location, m.dataframe, center=mean)
```

....

Levene's Test for Homogeneity of Variance (center = mean)

	Df	F value	Pr(>F)
group	1	7e-04	0.9799
	18		

....

```
t.test(L,C,var.equal=TRUE,alternative="less")
```

....

```
        Two Sample t-test

data:  L and C

t = -1.6341, df = 18, p-value = 0.0598

alternative hypothesis: true difference in means is less than 0

95 percent confidence interval:

      -Inf        0.1333561

sample estimates:

mean of x      mean of y

    16.70        18.88
```

....